

other assets contain many different factors. Corporate bonds, hedge funds, and private equity contain different amounts of equity risk, volatility risk, interest rate risk, and default risk. Factor theory predicts these assets have risk premiums that reflect their underlying factor risks.

3. Different investors need different risk factors.

Just as different people have different nutrient needs, different investors have different optimal exposures to different sets of risk factors.

Volatility, as we shall see, is an important factor. Many assets and strategies lose money during times of high volatility, such as observed during the 2007–2008 financial crisis. Most investors dislike these times and would prefer to be protected against large increases in volatility. A few brave investors can afford to take the opposite position; these investors can weather losses during bad times to collect a volatility premium during normal times. They are paid risk premiums as compensation for taking losses—sometimes big losses, as in 2008–2009—during volatile times.

Another example is that investors have different desired exposure to economic growth. One investor may not like times of shrinking GDP growth because he is likely to become unemployed in such circumstances. Another investor—a bankruptcy lawyer, perhaps—can tolerate low GDP growth because his labor income increases during recessions. The point is that each investor has different preferences, or risk aversion coefficients, for each different source of factor risk.

There is one difference, however, between factors and nutrients. Nutrients are inherently good for you. Factor risks are bad. It is by enduring these bad experiences that we are rewarded with risk premiums. Each different factor defines a different set of bad times. They can be bad economic times—like periods of high inflation and low economic growth. They can be bad times for investments—periods when the aggregate market or certain investment strategies perform badly. Investors exposed to losses during bad times are compensated by risk premiums in good times. The factor theory of investing specifies different types of underlying factor risk, where each different factor represents a different set of bad times or experiences. We describe the theory of factor risk by starting with the most basic factor risk premium theory—the CAPM, which specifies just one factor: the market portfolio.

3. CAPM

The CAPM was revolutionary because it was the first cogent theory to recognize that the risk of an asset was not how that asset behaved in isolation but how that asset moved in relation to other assets and to the market as a whole. Before the CAPM, risk was often thought to be an asset's own volatility. The CAPM said